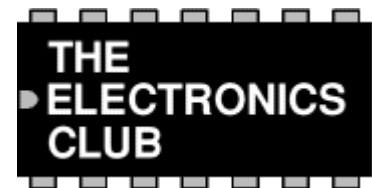


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Analogue and Digital Systems



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Analogue systems

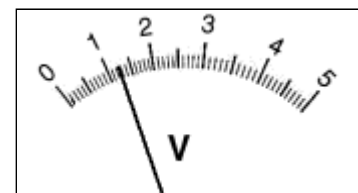
Analogue systems process analogue signals which can take any value within a range, for example the output from an [LDR](#) (light sensor) or a microphone.

An [audio amplifier](#) is an example of an analogue system. The amplifier produces an output voltage which can be any value within the range of its power supply.

An [analogue meter](#) can display any value within the range available on its scale. However, the precision of readings is limited by our ability to read them. For example the meter on the right shows 1.25V because the pointer is estimated to be half way between 1.2 and 1.3. The analogue meter can show any value between 1.2 and 1.3 but we are unable to read the scale more precisely than about half a division.



Analogue signal



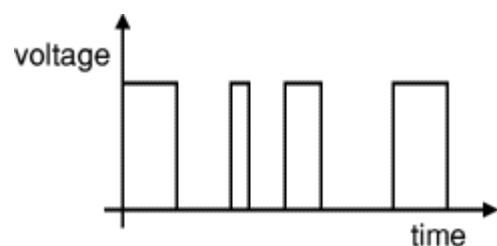
Analogue meter display

All electronic circuits suffer from 'noise' which is unwanted signal mixed in with the desired signal, for example an audio amplifier may pick up some mains 'hum' (the 50Hz frequency of the UK mains electricity supply). Noise can be difficult to eliminate from analogue signals because it may be hard to distinguish from the desired signal.

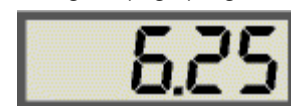
Digital systems

Digital systems process digital signals which can take only a limited number of values (discrete steps), usually just two values are used: the positive supply voltage (+Vs) and zero volts (0V).

Digital systems contain devices such as [logic gates](#), flip-flops, shift registers and counters. A computer is an example of a digital system.



Digital (logic) signal



Digital meter display

A [digital meter](#) can display many values, but not every value within its range. For example the display on the right can show 6.25 and 6.26 but not a value between them. This is not a problem because digital meters normally have sufficient digits to show values more precisely than it is possible to read an analogue display.

Logic signals

Most digital systems use the simplest possible type of signal which has just two values. This type of signal is called a **logic signal** because the two values (or states) can be called true and false. Normally the positive supply voltage +Vs represents true and 0V represents false. Other labels for the true and false states are shown in the table on the right.

Logic states	
True	False
1	0
High	Low
+Vs	0V
On	Off

Noise is relatively easy to eliminate from digital signals because it is easy to distinguish from the desired signal which can only have particular values. For example: if the signal is meant to be +5V (true) or 0V (false), noise of up to 2.5V can be eliminated by treating all voltages greater than 2.5V as true and all voltages less than 2.5V as false.

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